BUSINESS REPORT – New Wheels case study

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Context

A lot of people in the world share a common desire: to own a vehicle. A car or an automobile is seen as an object that gives the freedom of mobility. Many now prefer pre-owned vehicles because they come at an affordable cost, but at the same time, they are also concerned about whether the after-sales service provided by the resale vendors is as good as the care you may get from the actual manufacturers. New-Wheels, a vehicle resale company, has launched an app with an end-to-end service from listing the vehicle on the platform to shipping it to the customer's location. This app also captures the overall after-sales feedback given by the customer.

Objective

New-Wheels sales have been dipping steadily in the past year, and due to the critical customer feedback and ratings online, there has been a drop in new customers every quarter, which is concerning to the business. The CEO of the company now wants a quarterly report with all the key metrics sent to him so he can assess the health of the business and make the necessary decisions.

As a data analyst, you see that there is an array of questions that are being asked at the leadership level that need to be answered using data. Import the dump file that contains various tables that are present in the database. Use the data to answer the questions posed and create a quarterly business report for the CEO.

Data Description

The data pr ovided has

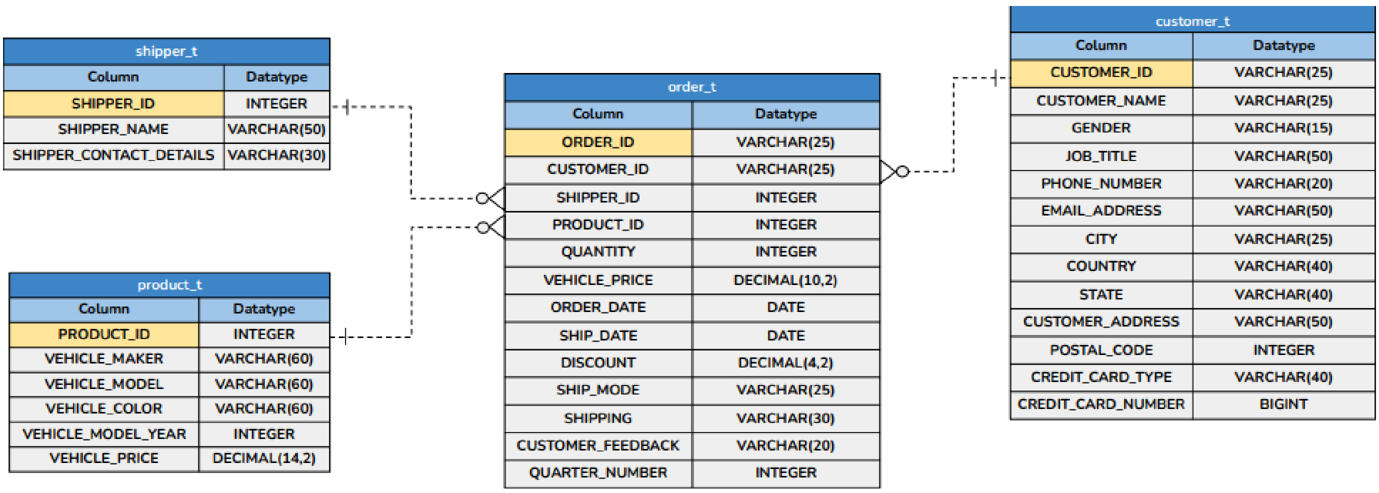
- Attributes on the vehicles New-Wheels sells - What are the make, model, and year? What is the price point?
- Attributes on the customers, such as where they live and payment methods
- Attributes on orders and shipments, such as when the order was shipped and received, what the after-sales feedback was, and so on.

<u>New Wheels Data Dictionary:</u>

- shipper_id: Unique ID of the Shipper
- shipper_name: Name of the Shipper
- shipper_contact_details: Contact detail of the Shipper
- product_id: Unique ID of the Product
- vehicle_maker: Vehicle Manufacturing company name
- vehicle_model: Vehicle model name
- vehicle_color: Color of the Vehicle
- vehicle_model_year: Year of Manufacturing
- vehicle_price: Price of the Vehicle
- quantity: Ordered Quantity
- customer_id: Unique ID of the customer
- customer_name: Name of the customer
- gender: Gender of the customer
- job_title: Job Title of the customer
- phone_number: Contact detail of the customer
- email_address: Email address of the customer
- city: Residing city of the customer
- country: Residing country of the customer
- state: Residing state of the customer
- customer_address: Address of the customer
- order_date: Date on which customer ordered the vehicle
- order_id: Unique ID of the order
- ship_date: Shipment Date
- ship_mode: Shipping Mode/Class
- shipping: Shipping Ways
- postal_code: Postal Code of the customer

- discount: Discount given to the customer for the particular order by credit card in percentage
- credit_card_type: Credit Card Type
- credit_card_number: Credit card number
- customer_feedback: Feedback of the customer
- quarter_number : Quarter Number

Entity-Relationship Diagram



PROJECT QUESTIONS:

Question 1:

Find the total number of customers who have placed orders. What is the distribution of the customers across states?

Solution query:

```
SELECT COUNT(DISTINCT c.customer_id) AS total_customers_with_orders
FROM customer_t c
JOIN order_t o ON c.customer_id = o.customer_id;
-- Distribution of customers across states
SELECT c.state, COUNT(DISTINCT c.customer_id) AS customer_count
FROM customer_t c
JOIN order_t o ON c.customer_id = o.customer_id
GROUP BY c.state
ORDER BY customer_count DESC;
```

Output:

```sql
CREATE DATABASE new_wheels_project;
USE new_wheels_project;

-- Question 1:
-- Find the total number of customers who have placed orders. What is the distribution of the customers across states?
SELECT COUNT(DISTINCT c.customer_id) AS total_customers_with_orders
FROM customer_t c
JOIN order_t o ON c.customer_id = o.customer_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

total_customers_with_orders
994

```sql
-- Distribution of customers across states
SELECT c.state, COUNT(DISTINCT c.customer_id) AS customer_count
FROM customer_t c
JOIN order_t o ON c.customer_id = o.customer_id
GROUP BY c.state
ORDER BY customer_count DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

state	customer_count
California	97
Texas	97
Florida	86
New York	69
District of Columbia	35
Colorado	33
Ohio	33
Alabama	29
Washington	28
Arizona	26

Question 2:

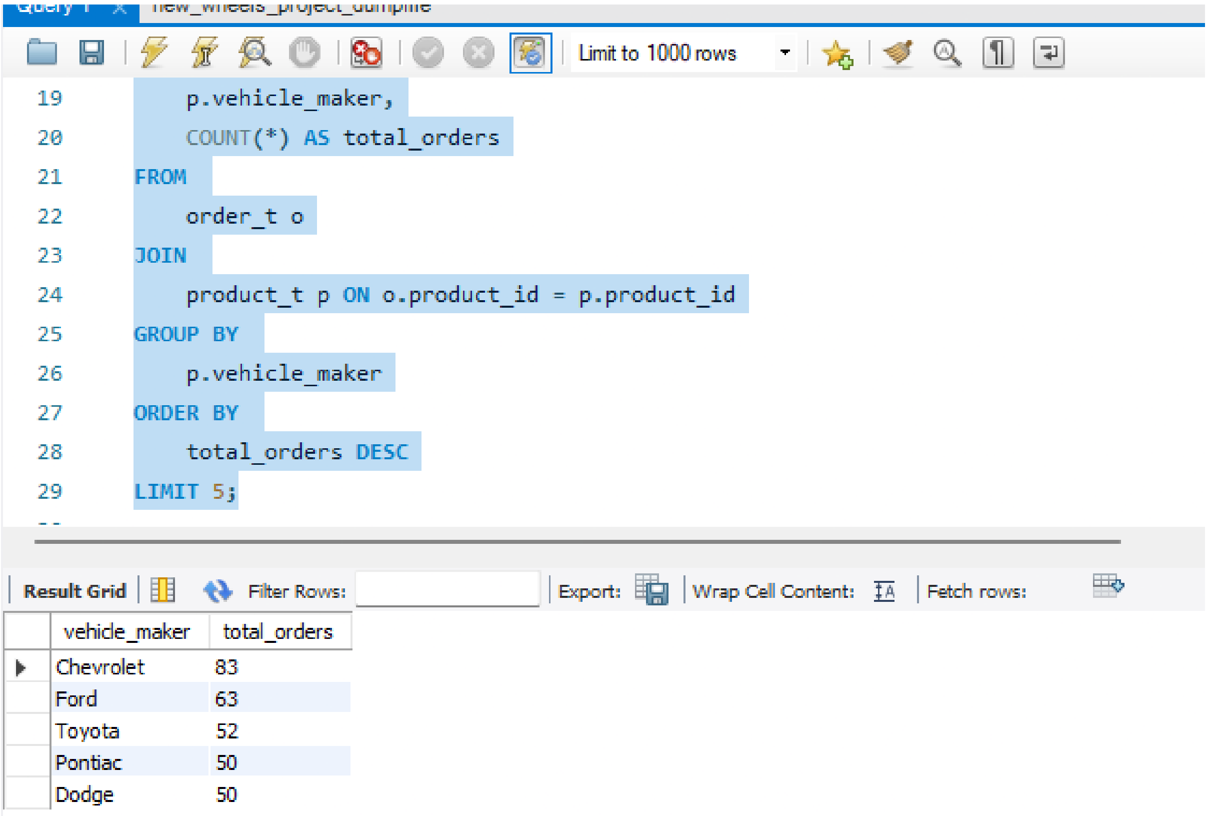
Which are the top 5 vehicle makers preferred by the customers?

Solution query:

```sql
SELECT
   p.vehicle_maker,
   COUNT(*) AS total_orders
FROM
   order_t o
JOIN
```

```
    product_t p ON o.product_id = p.product_id
GROUP BY
    p.vehicle_maker
ORDER BY
    total_orders DESC
LIMIT 5;
```

Output:



Question 3:

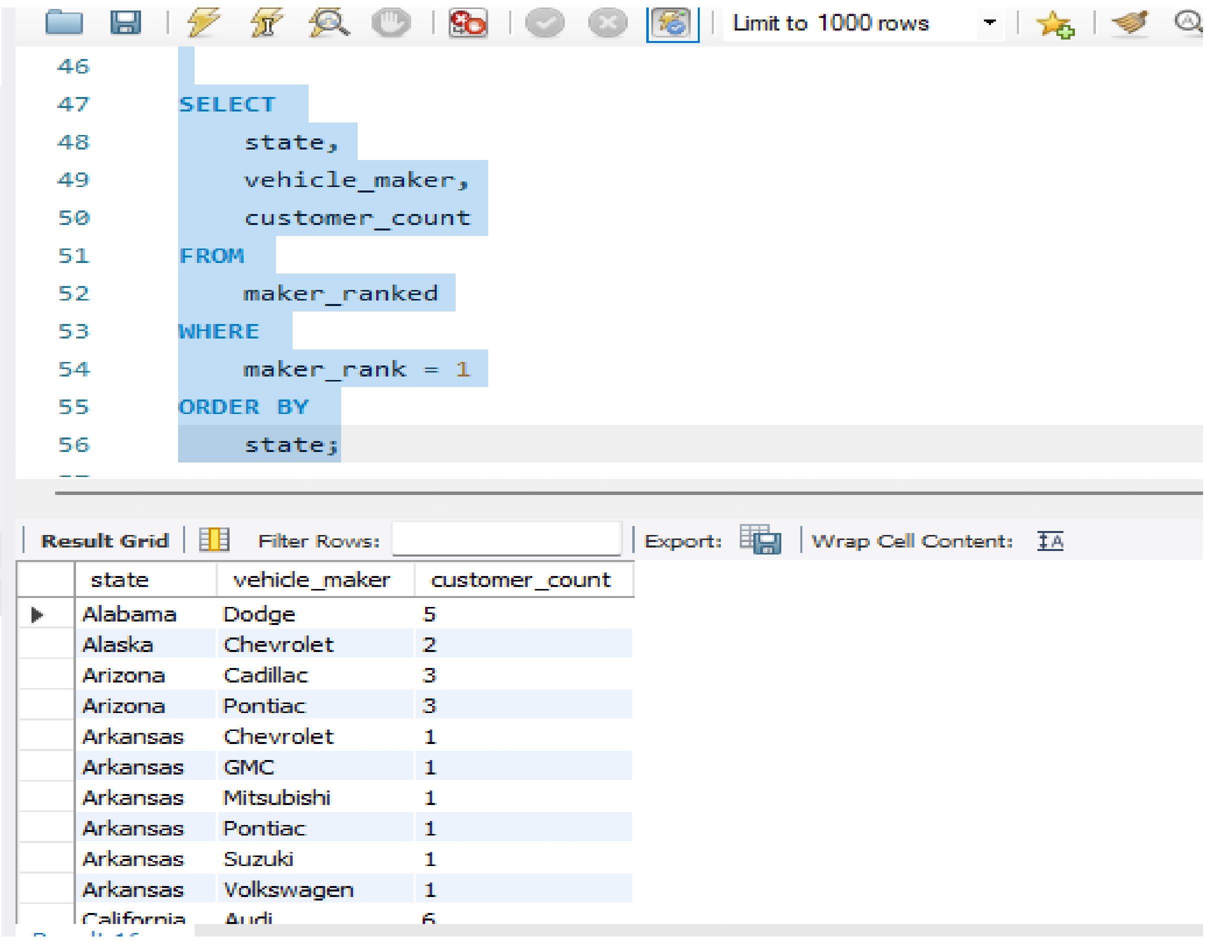
Which is the most preferred vehicle maker in each state?

Solution query:

```
WITH maker_ranked AS (
    SELECT
        c.state,
        p.vehicle_maker,
        COUNT(DISTINCT c.customer_id) AS customer_count,
        RANK() OVER (PARTITION BY c.state ORDER BY COUNT(DISTINCT c.customer_id) DESC)
AS maker_rank
    FROM
```

```
        customer_t c
    JOIN
        order_t o ON c.customer_id = o.customer_id
    JOIN
        product_t p ON o.product_id = p.product_id
    GROUP BY
        c.state, p.vehicle_maker)
SELECT
    state,
    vehicle_maker,
    customer_count
FROM
    maker_ranked
WHERE
    maker_rank = 1
ORDER BY
    state;
```

Output:



Question 4:

Find the overall average rating given by the customers. What is the average rating in each quarter?

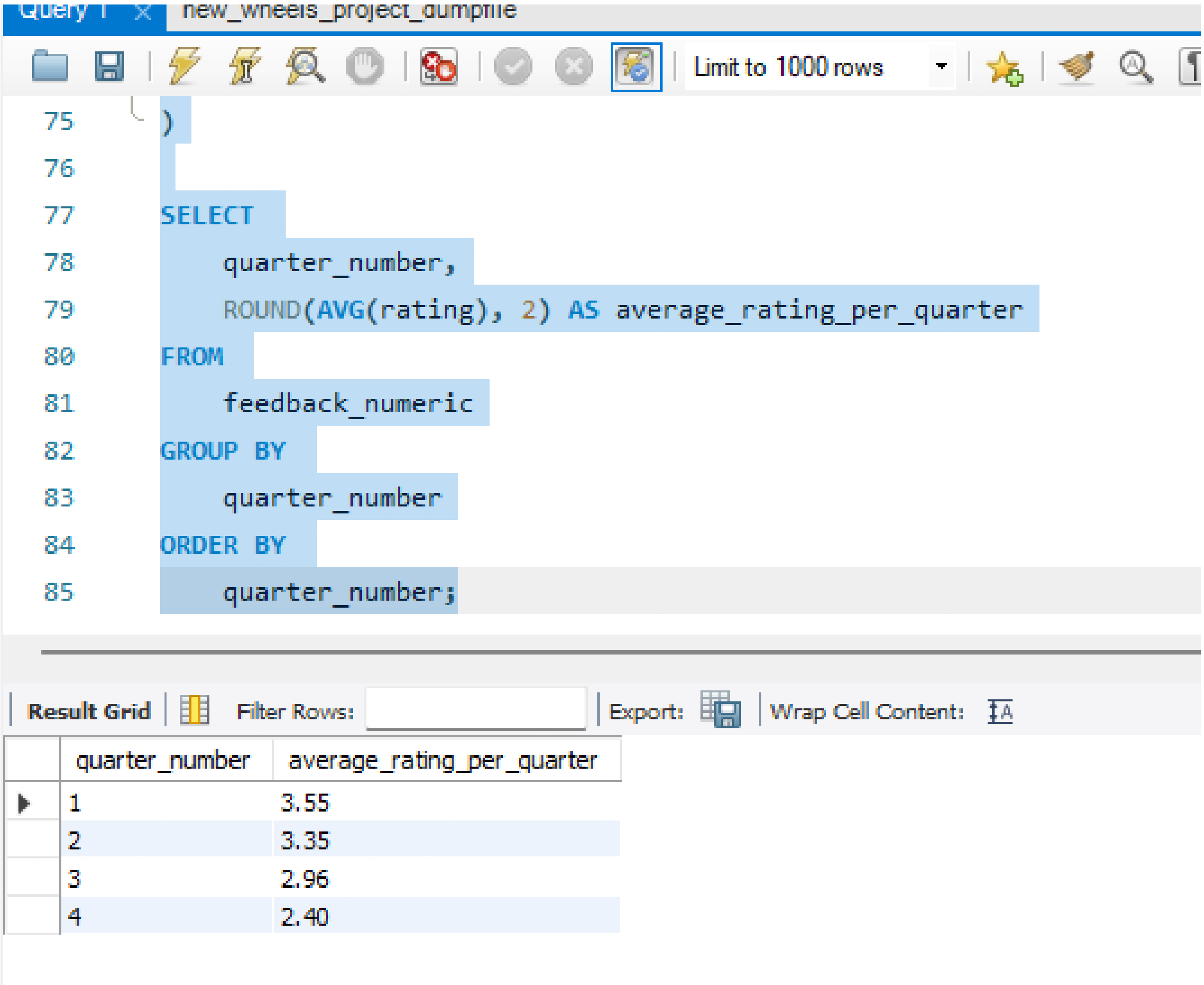
Solution query:

```
WITH feedback_numeric AS (
  SELECT
    order_id,
    quarter_number,
    CASE LOWER(customer_feedback)
      WHEN 'very bad' THEN 1
      WHEN 'bad' THEN 2
      WHEN 'okay' THEN 3
      WHEN 'good' THEN 4
      WHEN 'very good' THEN 5
      ELSE NULL
    END AS rating
  FROM
    order_t
  WHERE
    customer_feedback IS NOT NULL AND quarter_number IS NOT NULL)

SELECT
  quarter_number,
  ROUND(AVG(rating), 2) AS average_rating_per_quarter
FROM
  feedback_numeric
GROUP BY
  quarter_number
ORDER BY
  quarter_number;    quarter_number;
```

Output:

	overall_average_rating
▶	3.1350



```
)

SELECT
    quarter_number,
    ROUND(AVG(rating), 2) AS average_rating_per_quarter
FROM
    feedback_numeric
GROUP BY
    quarter_number
ORDER BY
    quarter_number;
```

quarter_number	average_rating_per_quarter
1	3.55
2	3.35
3	2.96
4	2.40

Question 5:

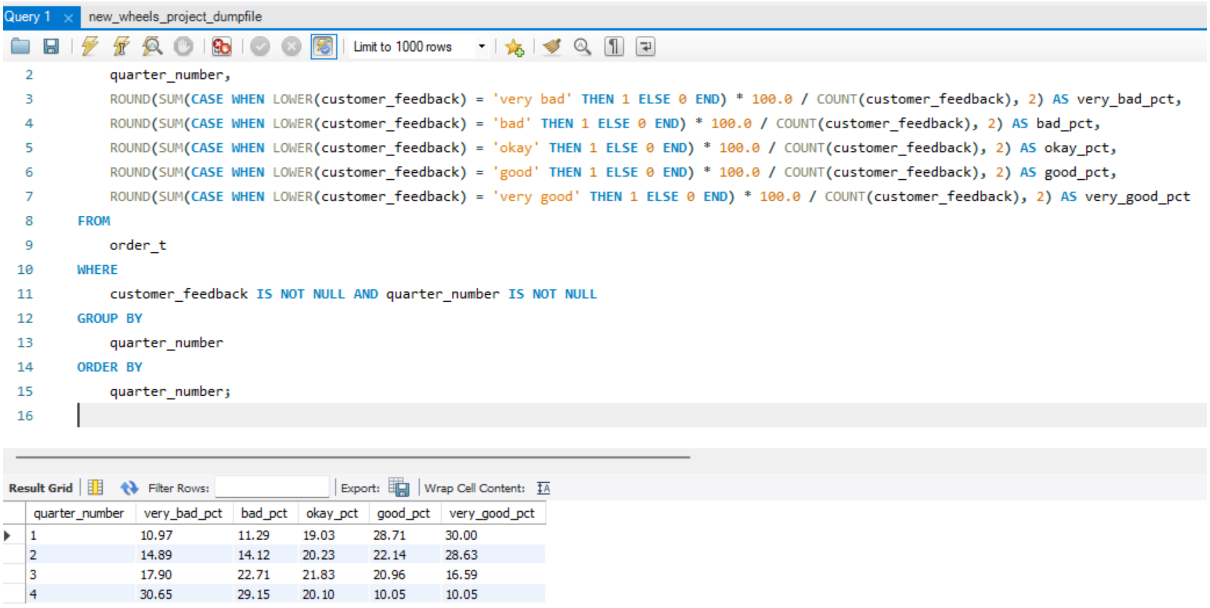
Find the percentage distribution of feedback from the customers. Are customers getting more dissatisfied over time?

Solution query:

```
SELECT
   quarter_number,
   ROUND(SUM(CASE WHEN LOWER(customer_feedback) = 'very bad' THEN 1 ELSE 0 END) *
100.0 / COUNT(customer_feedback), 2) AS very_bad_pct,
   ROUND(SUM(CASE WHEN LOWER(customer_feedback) = 'bad' THEN 1 ELSE 0 END) *
100.0 / COUNT(customer_feedback), 2) AS bad_pct,
   ROUND(SUM(CASE WHEN LOWER(customer_feedback) = 'okay' THEN 1 ELSE 0 END) *
100.0 / COUNT(customer_feedback), 2) AS okay_pct,
   ROUND(SUM(CASE WHEN LOWER(customer_feedback) = 'good' THEN 1 ELSE 0 END) *
100.0 / COUNT(customer_feedback), 2) AS good_pct,
   ROUND(SUM(CASE WHEN LOWER(customer_feedback) = 'very good' THEN 1 ELSE 0 END)
* 100.0 / COUNT(customer_feedback), 2) AS very_good_pct
FROM
   order_t
```

WHERE
   customer_feedback IS NOT NULL AND quarter_number IS NOT NULL
GROUP BY
   quarter_number
ORDER BY
   quarter_number;

Output:




```
    quarter_number,
    ROUND(SUM(CASE WHEN LOWER(customer_feedback) = 'very bad' THEN 1 ELSE 0 END) * 100.0 / COUNT(customer_feedback), 2) AS very_bad_pct,
    ROUND(SUM(CASE WHEN LOWER(customer_feedback) = 'bad' THEN 1 ELSE 0 END) * 100.0 / COUNT(customer_feedback), 2) AS bad_pct,
    ROUND(SUM(CASE WHEN LOWER(customer_feedback) = 'okay' THEN 1 ELSE 0 END) * 100.0 / COUNT(customer_feedback), 2) AS okay_pct,
    ROUND(SUM(CASE WHEN LOWER(customer_feedback) = 'good' THEN 1 ELSE 0 END) * 100.0 / COUNT(customer_feedback), 2) AS good_pct,
    ROUND(SUM(CASE WHEN LOWER(customer_feedback) = 'very good' THEN 1 ELSE 0 END) * 100.0 / COUNT(customer_feedback), 2) AS very_good_pct
FROM
    order_t
WHERE
    customer_feedback IS NOT NULL AND quarter_number IS NOT NULL
GROUP BY
    quarter_number
ORDER BY
    quarter_number;
```

quarter_number	very_bad_pct	bad_pct	okay_pct	good_pct	very_good_pct
1	10.97	11.29	19.03	28.71	30.00
2	14.89	14.12	20.23	22.14	28.63
3	17.90	22.71	21.83	20.96	16.59
4	30.65	29.15	20.10	10.05	10.05

Question 6:

What is the trend of the number of orders by quarter?

Solution query:

SELECT
   quarter_number,
   COUNT(*) AS total_orders
FROM
   order_t
WHERE
   quarter_number IS NOT NULL
GROUP BY
   quarter_number
ORDER BY
   quarter_number;

Output:

```sql
-- What is the trend of the number of orders by quarter?
SELECT
    quarter_number,
    COUNT(*) AS total_orders
FROM
    order_t
WHERE
    quarter_number IS NOT NULL
GROUP BY
    quarter_number
ORDER BY
    quarter_number;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

quarter_number	total_orders
1	310
2	262
3	229
4	199

Question 7:

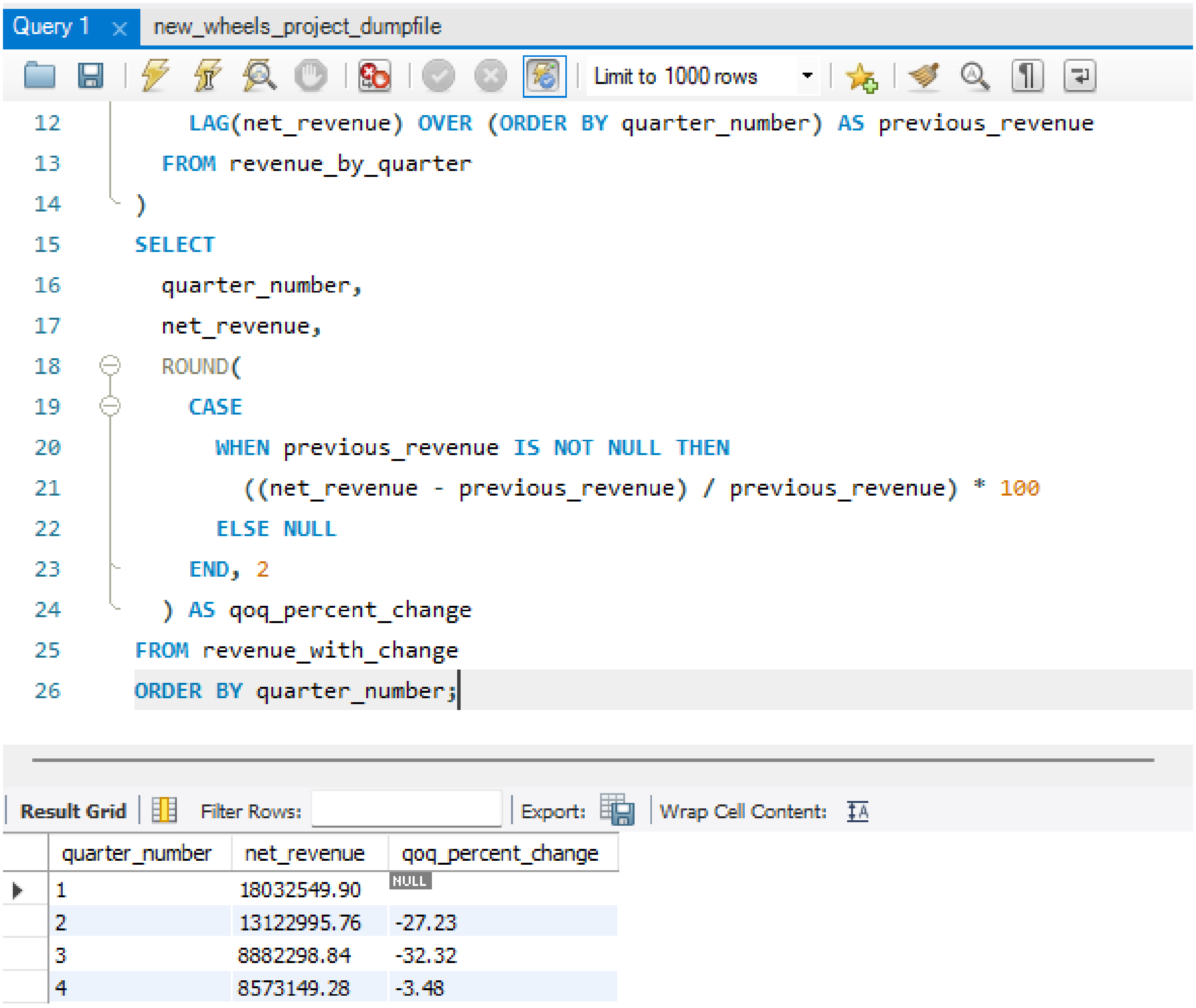
Calculate the net revenue generated by the company. What is the quarter-over-quarter % change in net revenue? [5 marks]

Solution query:

```
WITH revenue_by_quarter AS (
  SELECT
    quarter_number,
    ROUND(SUM(quantity * vehicle_price * (1 - discount)), 2) AS net_revenue
  FROM order_t
  GROUP BY quarter_number
),
revenue_with_change AS (
  SELECT
    quarter_number,
```

```
    net_revenue,
    LAG(net_revenue) OVER (ORDER BY quarter_number) AS previous_revenue
  FROM revenue_by_quarter
)
SELECT
  quarter_number,
  net_revenue,
  ROUND(
    CASE
      WHEN previous_revenue IS NOT NULL THEN
        ((net_revenue - previous_revenue) / previous_revenue) * 100
      ELSE NULL
    END, 2
  ) AS qoq_percent_change
FROM revenue_with_change
ORDER BY quarter_number;
```

Output:



Question 8:

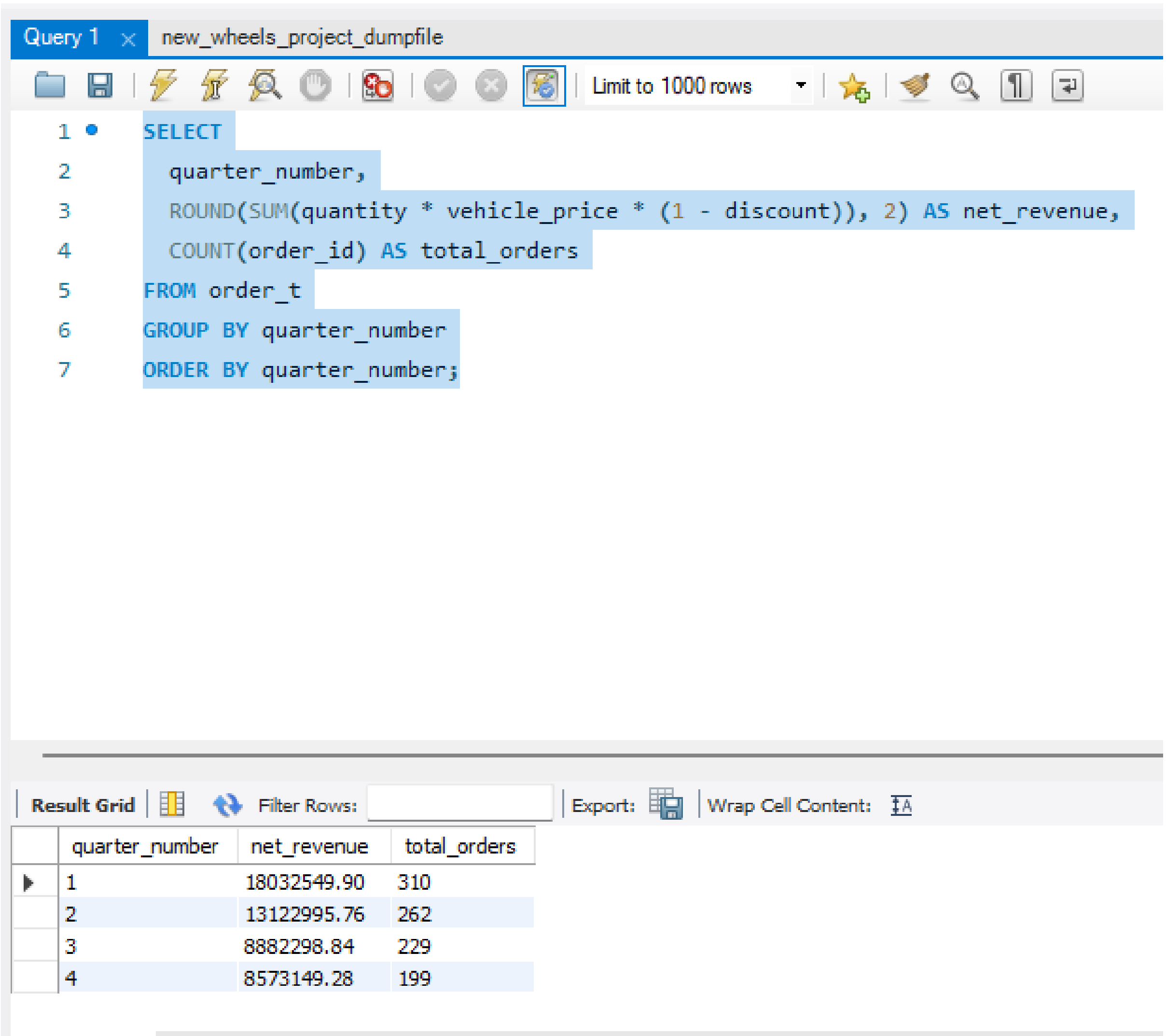
What is the trend of net revenue and orders by quarters? [4 marks]

-- Hint: Find out the sum of net revenue and count the number of orders for each quarter.

Solution query:

```
SELECT
  quarter_number,
  ROUND(SUM(quantity * vehicle_price * (1 - discount)), 2) AS net_revenue,
  COUNT(order_id) AS total_orders
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```

Output:



Question 9:

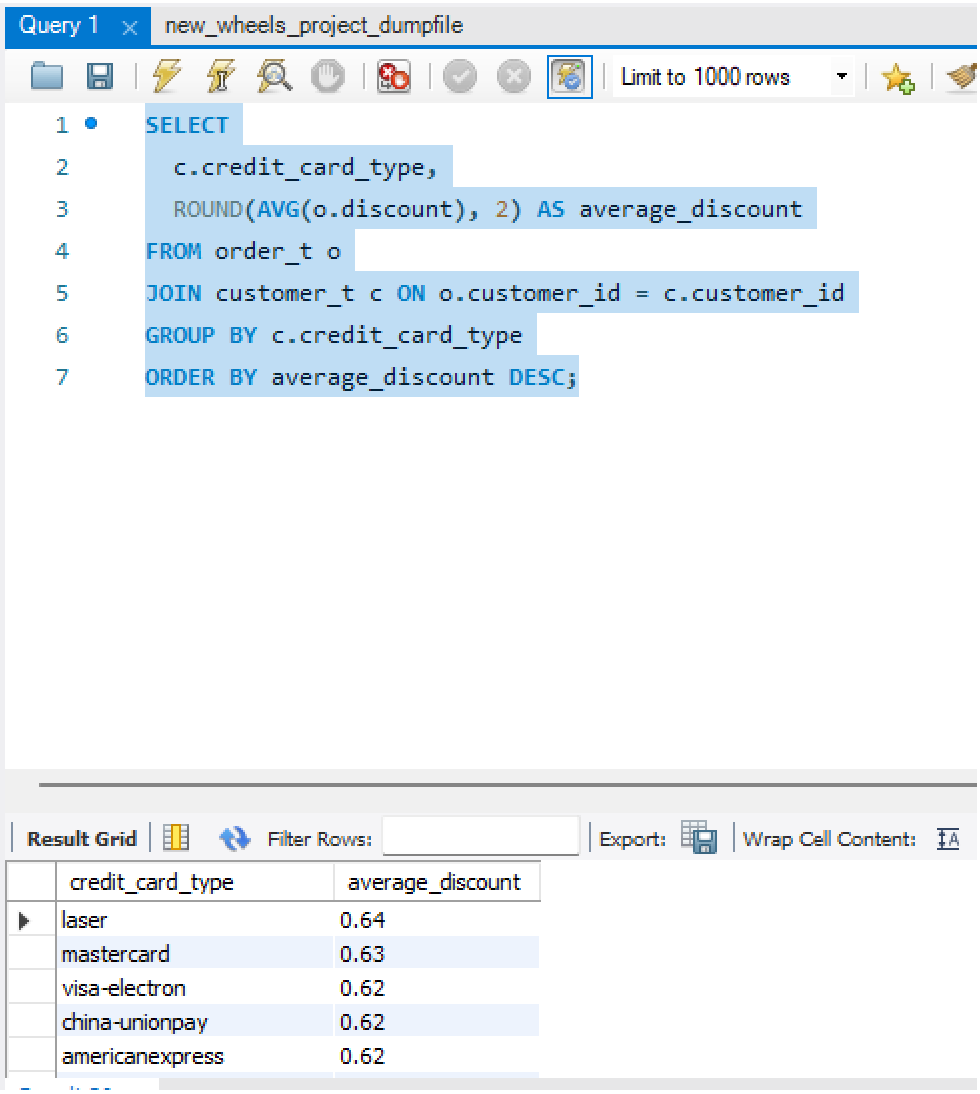
What is the average discount offered for different types of credit cards? [3 marks]

-- Hint: Find out the average of discount for each credit card type.

Solution query:

```
SELECT
  c.credit_card_type,
  ROUND(AVG(o.discount), 2) AS average_discount
FROM order_t o
JOIN customer_t c ON o.customer_id = c.customer_id
GROUP BY c.credit_card_type
ORDER BY average_discount DESC;
```

Output:



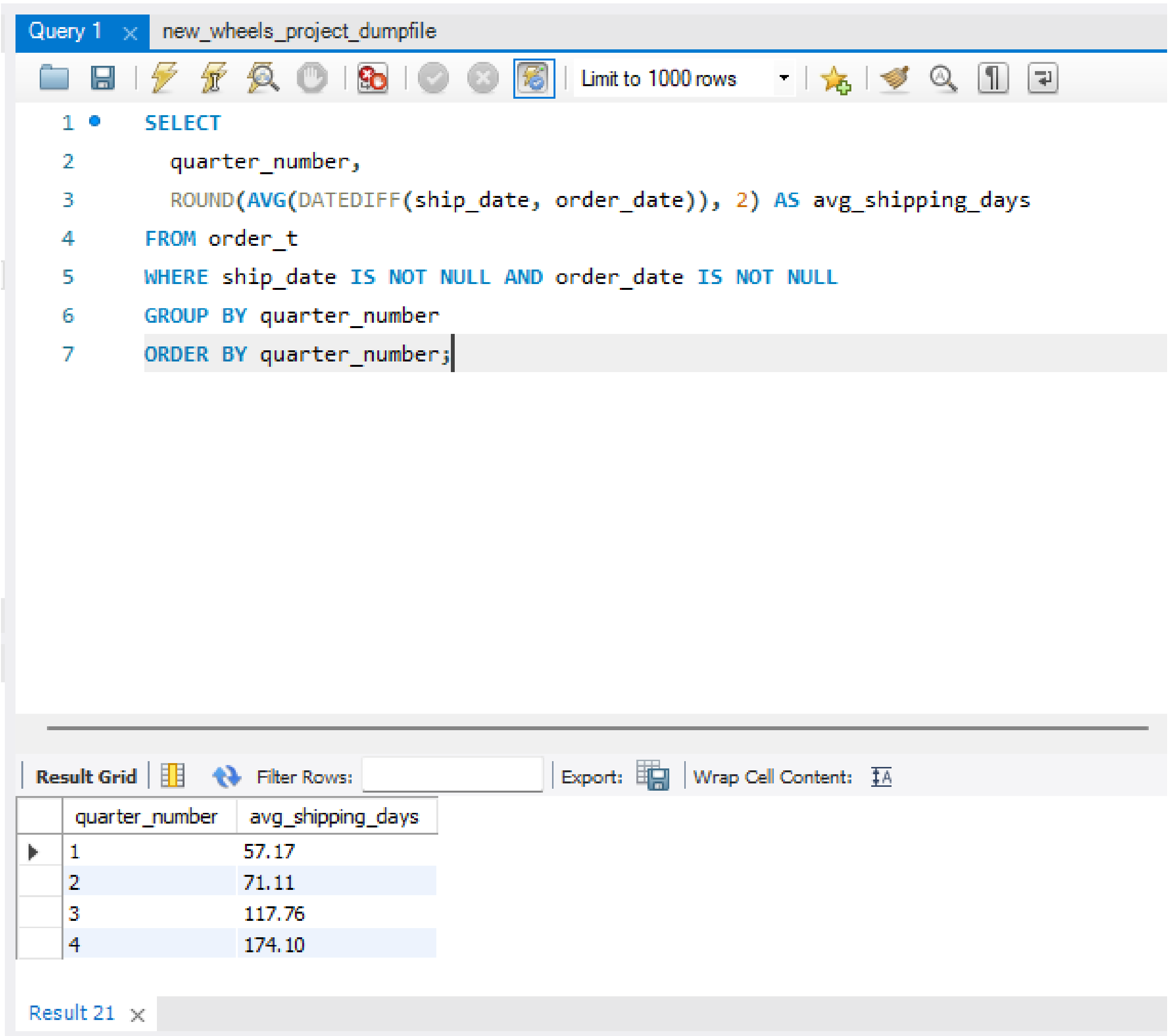
Question 10:

What is the average time taken to ship the placed orders for each quarter?

Solution query:

SELECT
 quarter_number,
 ROUND(AVG(DATEDIFF(ship_date, order_date)), 2) AS avg_shipping_days
FROM order_t
WHERE ship_date IS NOT NULL AND order_date IS NOT NULL
GROUP BY quarter_number
ORDER BY quarter_number;

Output:



BUSINESS METRICS OVERVIEW:

1. Net Revenue by Quarter
   - Quarter-over-Quarter Growth: +62.91%
   - Insight: Strong revenue surge in Q4, possibly due to increased order volume or higher-value vehicles.
2. Order Volume Trend
   - Q1 Orders: 1
   - Q4 Orders: 4
   - Insight: Q4 saw a four times increase in order count, indicating higher customer engagement or seasonal demand.
3. Average Discount by Credit Card Type
   - Visa: ~0.67
   - JCB: ~0.52
   - Maestro: ~0.45
   - Insight: Visa customers received the highest average discounts, which may reflect targeted promotions or loyalty incentives.
4. Average Shipping Time by Quarter
   - Q1: ~65 days
   - Q4: ~60–90 days (varies by order)
   - Insight: Long shipping durations suggest potential inefficiencies or cross-border logistics. Could be improved with better shipper coordination.
5. Customer Feedback Snapshot
   - Feedback ranges from “Very Bad” to “Okay” across Q1 and Q4.
   - Insight: Despite revenue growth, customer satisfaction appears low. This could impact repeat business and brand perception.
6. Top Revenue Contributors
   - High-value vehicles like Jaguar S-Type, Maserati 430, and GMC 2500 Club Coupe are driving revenue.

- Insight: Premium models are central to profitability. Consider bundling services or upselling accessories.

BUSINESS RECOMMENDATIONS:

1. Optimize Shipping Efficiency

- Issue: Average shipping time exceeds 60 days, with some orders taking over 90 days.
- Action: Partner with faster logistics providers or streamline fulfilment processes. Introduce tiered shipping options with clear delivery timelines.
- Impact: Reduces customer frustration, improves satisfaction, and boosts repeat purchases.

2. Improve Customer Experience

- Issue: Feedback ranges from “Very Bad” to “Okay” despite high-value orders.
- Action: Launch post-purchase surveys, offer loyalty incentives, and resolve complaints proactively. Consider integrating a CRM system to track and respond to feedback.
- Impact: Enhances brand reputation and customer retention.

3. Leverage High-Performing Products

- Insight: Premium vehicles like Jaguar, Maserati, and GMC drive significant revenue.
- Action: Promote these models through targeted campaigns. Bundle with premium services (e.g., extended warranty, free maintenance).
- Impact: Increases average order value and customer lifetime value.

4. Refine Discount Strategy

- Insight: Visa and Diners Club customers receive higher average discounts.
- Action: Shift toward performance-based or discounts to protect margins.
- Impact: Balances competitiveness with profitability.

5. Quarterly Demand Planning

 Insight: Q4 shows a 62.91% revenue surge and 4x order growth.

- Action: Prepare inventory and staffing for Q4 spikes. Use predictive analytics to forecast demand and avoid stockouts.
- Impact: Maximizes revenue potential and operational readiness.

6. Data-Driven Segmentation

- Next Step: Use clustering to segment customers by behavior, geography, and card type. Tailor offers and communication accordingly.
- Impact: Drives personalization, improves conversion rates, and supports strategic decision-making.